Uniform Sampling on the Sphere:
From Probability-Uniform Random Points to Quasi-Uniform Fibonacci Configurations

A Research Note for Convex Geometry, Support-Direction Discretisation, and Constant-Width Volume Computation

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Abstract

There are at least two mathematically distinct notions of "uniformity" in spherical sampling. The first is probabilistic: every random point has the normalised surface-area distribution. The second concerns the finite geometry of a configuration: points should neither cluster excessively nor leave large uncovered holes. This note develops that distinction in detail. We introduce separation radius, covering radius, mesh ratio, and quasi-uniformity; derive the natural $N^{-1/2}$ geometric scale on the two-dimensional sphere; and distinguish this scale from the $N^{-1/2}$ sampling error of the law of large numbers and the central limit theorem. We then derive the spherical Fibonacci construction from first principles: equal spacing in the height coordinate produces equal-area horizontal bands, the midpoint correction avoids polar degeneracy, and the golden-angle rotation suppresses low-order periodicity and near-resonance. The continued-fraction structure of the golden ratio is explained, together with its relation–and non-identity–to the 0.618 golden-section search in optimisation. Numerical illustrations compare i.i.d. uniform random points with Fibonacci configurations. Finally, the discussion is connected to the current dimension-raising computation for bodies of constant width: support directions, radial boundary directions, interior radial layers, upper-graph centre clouds, and the final i.i.d. Monte Carlo directions on $\mathbb{S}^3$. Convex geometry provides the application context; the central subject is spherical sampling itself.

Contents

1 Motivation: one word "uniform", two different requirements

In the present four-dimensional volume computation for bodies of constant width, spherical point sets are used for two different tasks. First, they describe geometry: many directions on $\mathbb{S}^2$ are used to approximate support constraints, radial boundaries, and the upper graph in the dimension-raising construction. Second, they compute an integral: random directions on $\mathbb{S}^3$ are used in a radial Monte Carlo estimator for four-dimensional volume.

Both operations may be described informally as "sampling the sphere uniformly", but their mathematical requirements are different.

Central distinction

Random points are uniform in statistical distribution,
Fibonacci points are more regular as a finite geometric layout.

Independent uniform points have the correct marginal distribution and support the law of large numbers. A finite random sample, however, is allowed to contain tight clusters and unusually large holes. A Fibonacci configuration is deterministic; its purpose is not independence, but more balanced separation, coverage, and area allocation at a fixed value of N.

1.1 Surface measure and probability-uniform sampling

Let

$$\mathbb{S}^2 = \{x \in \mathbb{R}^3 : \|x\| = 1\}$$

with standard area measure σ, so $\sigma(\mathbb{S}^2) = 4\pi$. The normalised probability measure is

$$\mu(A) = \frac{\sigma(A)}{4\pi}.$$

A random vector U is uniform on $\mathbb{S}^2$ if

$$\mathbb{P}(U \in A) = \mu(A)$$

for every measurable $A \subset \mathbb{S}^2$.

For i.i.d. uniform random variables $U_1, U_2, \ldots$, the law of large numbers gives, for any fixed measurable set A,

$$\frac{1}{N}\sum_{j=1}^{N} \mathbf{1}_A(U_j) \longrightarrow \mu(A) \quad \text{almost surely.}$$

More generally,

$$\frac{1}{N}\sum_{j=1}^{N} f(U_j) \longrightarrow \int_{\mathbb{S}^2} f \, d\mu$$

for integrable f. This is the probabilistic basis of ordinary Monte Carlo. Chapters 2 and 3 of Durrett provide a rigorous treatment of laws of large numbers and central limit theorems[3].

These statements concern empirical measures and averages. They do not assert that every finite realisation has regular nearest-neighbour distances or good worst-case coverage. Figure 1 compares an i.i.d. uniform sample and a spherical Fibonacci configuration with the same number of points.

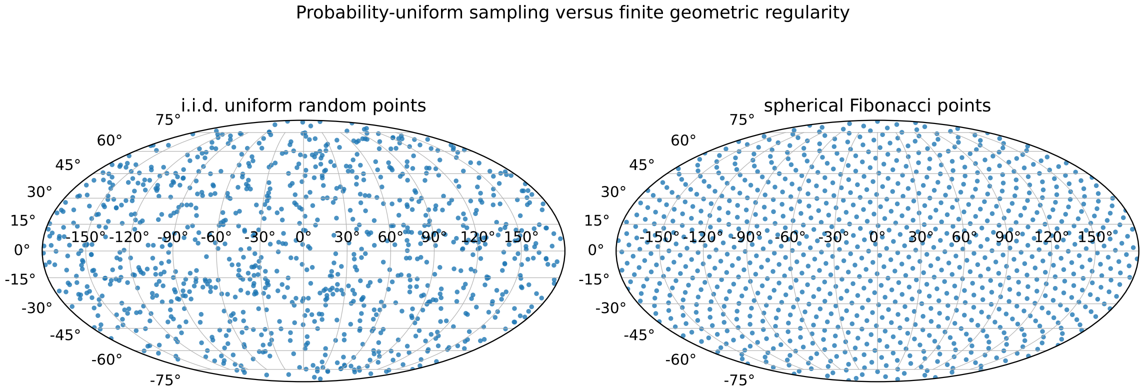

Figure 1: Probability-uniform sampling versus finite geometric regularity. The points on the left are genuinely i.i.d. uniform, but finite clustering and holes remain visible. The Fibonacci configuration on the right is deterministic and spatially more regular.

1.2 Finite geometric uniformity

For a finite configuration, the natural questions are different:

(1) Are any two points unnecessarily close?

(2) Is there a location on the sphere far from every sample point?

(3) As N increases, do clustering and hole scales shrink at comparable rates?

These are measured by separation radius, covering radius, and mesh ratio. They are standard quantities in discrete geometry, scattered-data approximation, spherical quadrature, and convex-body approximation[4, 5].

2 Three geometric quantities for finite spherical configurations

Let

$$X_N = \{x_1, \dots, x_N\} \subset \mathbb{S}^2.$$

We use the geodesic distance

$$d_{\mathbb{S}^2}(x, y) = \arccos\langle x, y\rangle.$$

For nearby points, chordal and geodesic distances are equivalent because

$$\|x - y\| = 2 \sin \frac{d_{\mathbb{S}^2}(x, y)}{2} = d_{\mathbb{S}^2}(x, y) + O(d_{\mathbb{S}^2}(x, y)^3).$$

Thus either metric gives the same asymptotic powers, although constants differ.

2.1 Separation radius: how tightly can points cluster?

Definition 2.1 (Separation radius)**.**

$$q(X_N) = \frac{1}{2} \min_{i \neq j} d_{\mathbb{S}^2}(x_i, x_j).$$

It is controlled by the closest pair. If $q(X_N) = 0.001$, then some pair is only 0.002 radians apart. The factor $1/2$ has a geometric purpose: the open spherical discs $B_{\mathbb{S}^2}(x_i, q)$ have disjoint interiors. If two such discs intersected, the triangle inequality would force two centres to be closer than $2q$, contradicting the definition.

> $\boxed{q(X_N) \text{ measures how crowded the configuration is.}}$
>
> A larger q means better separation. A very small q signals near-duplicate information.

2.2 Covering radius: how large is the largest hole?

Definition 2.2 (Covering radius or fill distance)**.**

$$h(X_N) = \sup_{x \in \mathbb{S}^2} \min_{1 \leq i \leq N} d_{\mathbb{S}^2}(x, x_i).$$

For each x, first find the nearest sample point. Then take the worst x on the sphere. Therefore $h(X_N)$ is the distance from the centre of the largest uncovered hole to the nearest node. Equivalently, it is the smallest h such that

$$\mathbb{S}^2 \subseteq \bigcup_{i=1}^{N} B_{\mathbb{S}^2}(x_i, h).$$

> $\boxed{h(X_N) \text{ measures whether the configuration leaves a large hole.}}$
>
> A smaller h means better worst-case coverage.

2.3 Mesh ratio: are holes and spacing on the same scale?

Definition 2.3 (Mesh ratio)**.**

$$\eta(X_N) = \frac{h(X_N)}{q(X_N)}.$$

The numerator is the largest-hole scale; the denominator is the closest-pair scale. Hence the mesh ratio does not measure absolute fineness. It measures whether global coverage and local spacing live on comparable scales.

Example 2.4 (Equal coverage, very different efficiency)**.** *Suppose two configurations both have* $N = 1000$ *points.*

For configuration A,

$$h = 0.12, \qquad q = 0.10,$$

so $\eta = 1.2$. *The largest hole and the separation scale are comparable.*

For configuration B,

$$h = 0.12, \qquad q = 0.0001,$$

so $\eta = 1200$. *The largest hole is no smaller, but at least two points are almost coincident. Many points are geometrically wasted.*

A large mesh ratio usually indicates severe local clustering relative to the global resolution.

2.4 Why not study h alone? Why not q alone?

The covering radius alone cannot detect redundant points. One could first use 500 well-spread points and then place another 500 almost on top of the north pole. The first 500 may still control h, so the covering radius does not reveal that half the computational budget was wasted. The separation radius becomes nearly zero, and h/q exposes the degeneration.

Conversely, a large separation radius does not guarantee global coverage. Points can be well separated but confined to the northern hemisphere, leaving the southern hemisphere empty. Thus

$$\boxed{h \text{ controls holes}, \qquad q \text{ controls clustering.}}$$

The ratio tests whether those two scales are compatible.

2.5 Why is the mesh ratio dimensionless?

Both h and q have units of angle or length, so h/q has no units. For a sensible sequence of refinements,

$$h \to 0, \qquad q \to 0.$$

Absolute scales must decrease as the mesh is refined. The meaningful question is whether the shape of the point pattern degenerates. The ratio removes the global scale, just as an aspect ratio describes the shape of a rectangle independently of its size.

3 Quasi-uniformity and the necessity of an N-independent constant

Definition 3.1 (Quasi-uniform sequence). *A sequence of configurations* $X_N \subset \mathbb{S}^2$ *is quasi-uniform if there is a constant* $C < \infty$*, independent of* N*, such that*

$$\frac{h(X_N)}{q(X_N)} \le C$$

for all sufficiently large N.

3.1 Why must C be independent of N?

We are studying an entire refinement sequence

$$X_1, X_2, \dots, X_N, \dots,$$

not a single finite set. If dependence on N were allowed, every configuration without repeated points would satisfy the vacuous statement

$$\eta(X_N) \le C_N, \qquad C_N := \eta(X_N).$$

For example, a sequence with $\eta(X_N) = N$ would be called "bounded" by taking $C_N = N$, even though its relative geometric quality becomes worse at every refinement.

A fixed constant means that the largest hole is never more than a fixed multiple of the closest-pair scale, whether N is one thousand, ten thousand, or one million. The mesh becomes finer without becoming more distorted.

Uniform shape regularity

$\boxed{C \text{ must work at every resolution.}}$

This is the point-cloud analogue of shape regularity in finite-element meshes: elements may shrink, but their shapes must not collapse.

4 Asymptotic notation and natural geometric scales

4.1 Meaning of $\asymp$

The notation

$$a_N \asymp b_N$$

means that there exist constants $c, C > 0$, independent of N, such that

$$cb_N \le a_N \le Cb_N$$

for all sufficiently large N. Hence

$$h(X_N) \asymp N^{-1/2}$$

means

$$cN^{-1/2} \le h(X_N) \le CN^{-1/2}.$$

The two quantities have the same order of magnitude.

This is weaker than asymptotic equivalence

$$a_N \sim b_N, \qquad \frac{a_N}{b_N} \to 1.$$

For example,

$$2N^{-1/2} \asymp N^{-1/2}$$

but not $2N^{-1/2} \sim N^{-1/2}$, whereas

$$N^{-1/2} + N^{-1} \sim N^{-1/2}.$$

It is also stronger than a one-sided big-O estimate, which gives only an upper bound.

4.2 Why is the natural scale on $\mathbb{S}^2$ equal to $N^{-1/2}$?

The essential fact is that $\mathbb{S}^2$ is a two-dimensional surface. Its total area is 4π. If N nodes divide the sphere approximately evenly, each node is responsible for area

$$\frac{4\pi}{N}.$$

Area is proportional to length squared at small scales. If ℓ_N is the corresponding local length scale, then

$$\ell_N^2 \asymp \frac{1}{N}, \qquad \boxed{\ell_N \asymp N^{-1/2}.}$$

The same conclusion is obvious on the unit square: if $N = m^2$ nodes form an $m \times m$ grid, the spacing is $1/m = N^{-1/2}$. Curvature changes constants, not the exponent.

4.3 Packing: q cannot be uniformly larger than $N^{-1/2}$

By definition, the N discs of radius q centred at the nodes are disjoint. A spherical cap of geodesic radius r has area

$$A(r) = 2\pi(1 - \cos r) = \pi r^2 + O(r^4).$$

Thus for small q each disc has area at least cq^2. Since all discs fit in area 4π,

$$Ncq^2 \leq 4\pi,$$

so

$$q \leq CN^{-1/2}.$$

No configuration can keep all nearest-neighbour distances on a scale substantially larger than $N^{-1/2}$.

4.4 Covering: h cannot be uniformly smaller than $N^{-1/2}$

The covering property says

$$\mathbb{S}^2 \subseteq \bigcup_{i=1}^{N} B_{\mathbb{S}^2}(x_i, h).$$

Each small cap has area at most Ch^2, and N such caps must cover area 4π. Hence

$$NCh^2 \geq 4\pi, \qquad h \geq cN^{-1/2}.$$

The largest-hole scale cannot be much smaller than $N^{-1/2}$.

4.5 Quasi-uniformity yields optimal orders for both h and q

The general geometric inequalities are

$$q \leq C_1 N^{-1/2}, \qquad h \geq c_1 N^{-1/2}.$$

If quasi-uniformity gives $h \le Cq$, then

$$c_1 N^{-1/2} \le h \le Cq \le CC_1 N^{-1/2},$$

so

$$h \asymp N^{-1/2}.$$

Also,

$$q \ge \frac{h}{C} \ge \frac{c_1}{C} N^{-1/2},$$

and therefore

$$q \asymp N^{-1/2}.$$

A bounded mesh ratio places both h and q on the optimal $N^{-1/2}$ scale.

4.6 General dimension

On a compact d-dimensional Riemannian manifold, small geodesic balls satisfy

$$\mathrm{Vol}_d(B(x,r)) \asymp r^d.$$

The same packing and covering arguments give the natural scale

$$h, q \asymp N^{-1/d}.$$

Thus

space	natural geometric spacing
$\mathbb{S}^1$	N^{-1}
$\mathbb{S}^2$	$N^{-1/2}$
$\mathbb{S}^3$	$N^{-1/3}$
d-dimensional compact manifold	$N^{-1/d}$

5 Why this $N^{-1/2}$ is not the law-of-large-numbers rate

The exponents coincide on $\mathbb{S}^2$, but the mechanisms are different.

5.1 The Monte Carlo rate

For i.i.d. variables with finite variance,

$$\mathrm{Var}\left(\frac{1}{N}\sum_{j=1}^{N} Y_j\right) = \frac{\mathrm{Var}(Y_1)}{N}.$$

The standard deviation of the sample mean is therefore proportional to $N^{-1/2}$. This comes from adding variances of independent random variables and is independent of the dimension of the integration domain.

5.2 The geometric rate

The relation

$$h, q \asymp N^{-1/2}$$

on $\mathbb{S}^2$ comes from

$$N \times (\text{length})^2 \asymp 1.$$

On $\mathbb{S}^3$ the geometric spacing is $N^{-1/3}$, whereas ordinary Monte Carlo error remains $N^{-1/2}$. Hence the two rates cannot have the same origin.

Same exponent - different causes

$$\boxed{\begin{aligned} N^{-1/2} \text{ in Monte Carlo} &: \text{ variance addition for independent samples,} \\ N^{-1/2} \text{ in } \mathbb{S}^2 \text{ geometry} &: \text{ two-dimensional area scales like length squared.} \end{aligned}}$$

5.3 Why random configurations are generally not quasi-uniform

Independent random points do not repel one another. A new point may fall arbitrarily close to an existing point, while extreme-value effects create holes larger than the average cell scale. Detailed separation and covering asymptotics for random spherical points are studied by Brauchart et al.[6].

A useful heuristic on $\mathbb{S}^2$ is:

- there are order N^2 pairs, and the probability that a pair is within distance r is order r^2, so the closest-pair scale is around N^{-1};
- covering every region introduces an extreme-value logarithm, leading to a largest-hole scale around $\sqrt{\log N/N}$.

Thus the random mesh ratio grows with N rather than remaining bounded. This does not contradict equidistribution of the empirical measure. It shows only that statistical uniformity does not imply finite geometric quasi-uniformity.

6 The spherical Fibonacci construction

The current program uses

Listing 1: Spherical Fibonacci generator in the project code

```
def fibonacci_sphere(n: int) -> np.ndarray:
    i = np.arange(n, dtype=float)
    z = 1.0 - 2.0 * (i + 0.5) / n
    phi = np.pi * (3.0 - np.sqrt(5.0)) * i
    r = np.sqrt(np.maximum(0.0, 1.0 - z * z))
    return np.column_stack((r * np.cos(phi),
                            r * np.sin(phi), z))
```

In mathematical notation,

$$z_i = 1 - \frac{2(i+1/2)}{N}, \qquad \phi_i = i\gamma, \qquad \gamma = \pi(3-\sqrt{5}),$$

and

$$x_i = \left(\sqrt{1-z_i^2}\cos\phi_i, \sqrt{1-z_i^2}\sin\phi_i, z_i\right).$$

Clearly $\|x_i\| = 1$. The two coordinates solve two distinct problems:

z_i creates equal-area north–south allocation,	ϕ_i suppresses longitudinal stripes and short periods.

7 Why equally spaced heights produce equal-area bands

7.1 Parameterising the sphere by height and longitude

Write

$$X(z,\phi) = \left(\sqrt{1-z^2}\cos\phi, \sqrt{1-z^2}\sin\phi, z\right),$$

for $-1 < z < 1$ and $0 \le \phi < 2\pi$. Then

$$X_z = \left(-\frac{z}{\sqrt{1-z^2}}\cos\phi, -\frac{z}{\sqrt{1-z^2}}\sin\phi, 1\right),$$

$$X_\phi = \left(-\sqrt{1-z^2}\sin\phi, \sqrt{1-z^2}\cos\phi, 0\right).$$

A direct calculation gives

$$\|X_z \times X_\phi\| = 1,$$

so the surface-area element is

$$\boxed{dA = dz\, d\phi.}$$

Consequently, the complete horizontal zone between heights z_1 and z_2 has area

$$\text{Area} = \int_{z_1}^{z_2}\int_0^{2\pi} d\phi\, dz = 2\pi(z_2 - z_1).$$

This is the coordinate form of Archimedes' spherical-zone theorem: the area depends only on the vertical height of the zone.

7.2 Equal intervals in z are equal areas on the sphere

Split $[-1,1]$ into N intervals of length $2/N$. Each corresponding spherical zone has area

$$2\pi\frac{2}{N} = \frac{4\pi}{N}.$$

The midpoint of the ith interval is exactly

$$z_i = 1 - \frac{2(i+1/2)}{N}.$$

Thus the construction divides the sphere into N equal-area horizontal zones and places one point at the middle height of each zone.

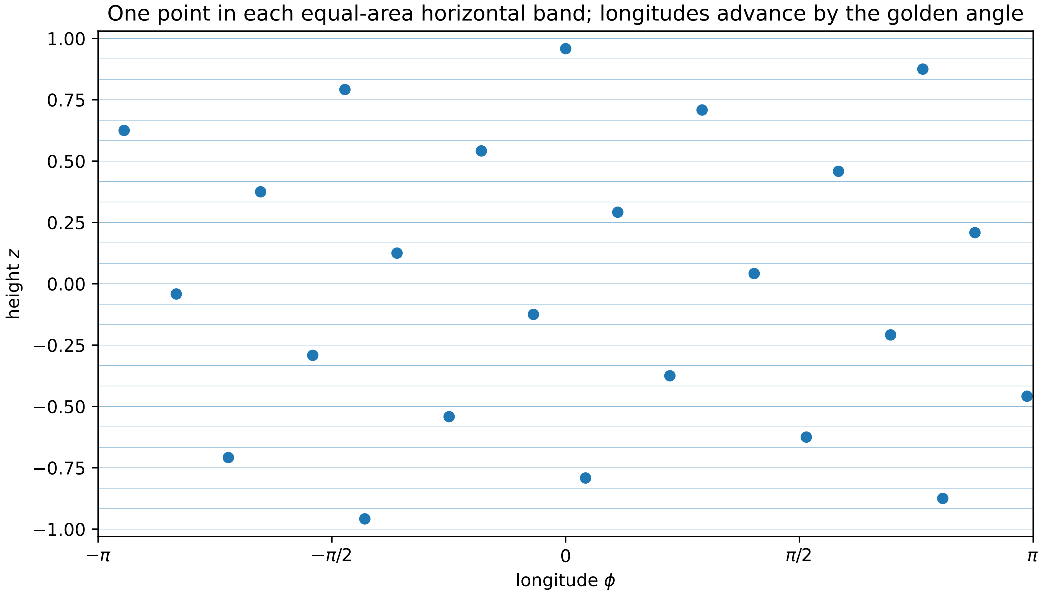

Figure 2: One point per equal-area horizontal zone. In (ϕ, z) coordinates the area element is simply $dz\, d\phi$; the longitudes advance by the golden angle.

7.3 Why the midpoint correction $i + 1/2$?

Without the half-step, $z_0 = 1$ would place a node exactly at the north pole, where longitude is degenerate. Including both endpoints also creates an asymmetry or duplicate polar handling. The midpoint choice avoids both poles, centres each node in its equal-area zone, and yields the symmetric extreme heights

$$1 - \frac{1}{N}, \qquad -1 + \frac{1}{N}.$$

7.4 Why not use equally spaced colatitudes θ?

With $z = \cos\theta$, the area element is

$$dA = \sin\theta\, d\theta\, d\phi.$$

Equal increments of θ do not produce equal-area bands: polar zones have much smaller circumference than equatorial zones. Placing the same number of nodes on equally spaced latitude levels therefore over-samples the poles. Since

$$dz = -\sin\theta\, d\theta,$$

the height coordinate absorbs the Jacobian factor. Hence

$$\boxed{\theta \text{ uniform is not area-uniform}, \qquad z = \cos\theta \text{ uniform is area-uniform.}}$$

Probabilistically, the third coordinate of a uniform point on $\mathbb{S}^2$ is uniform on $[-1, 1]$, while the colatitude has density proportional to $\sin\theta$.

8 Why the golden angle is needed

Equal-area heights alone are insufficient. If all ϕ_i were zero, every point would lie on one meridian. Each horizontal zone would contain a point, but most longitudes would be empty. The longitude sequence must avoid periodic stripes.

8.1 Algebra of the golden angle

Let

$$\varphi = \frac{1+\sqrt{5}}{2} \approx 1.6180339887.$$

Then

$$\varphi^2 = \varphi + 1, \qquad \frac{1}{\varphi} = \varphi - 1 \approx 0.618034,$$

$$\frac{1}{\varphi^2} = 1 - \frac{1}{\varphi} = \frac{3-\sqrt{5}}{2} \approx 0.381966.$$

The golden angle is the fraction $1/\varphi^2$ of a full turn:

$$\gamma = \frac{2\pi}{\varphi^2} = \pi(3-\sqrt{5}) \approx 2.399963 \text{ rad} \approx 137.5078^\circ.$$

The longitude is

$$\phi_i = i\gamma \pmod{2\pi}.$$

8.2 Why rational rotations create stripes

Normalise the circle to length 1 and set $\alpha = \gamma/(2\pi)$. If $\alpha = p/q$ is rational, then

$$q\alpha = p \equiv 0 \pmod{1}.$$

After q steps the sequence returns exactly to its starting longitude and uses at most q distinct meridians. A visible periodic pattern is unavoidable.

If α is irrational, exact repetition is impossible. More strongly, the equidistribution theorem for irrational rotations states that

$$\{i\alpha\}_{i\geq 0}$$

is uniformly distributed around the circle: the proportion of iterates in any arc converges to the arc-length proportion[9, 10]. Thus any irrational angle gives long-run equidistribution.

8.3 Why the golden ratio rather than an arbitrary irrational?

Finite prefixes are sensitive to near-periodicity. If

$$\alpha \approx \frac{p}{q}$$

with small q, then after q rotations the sequence nearly returns to its previous longitude. Such a good small-denominator rational approximation produces short-range stripes and redundant directions.

The golden ratio has the continued fraction

$$\varphi = [1; 1, 1, 1, \ldots].$$

All partial quotients are the smallest possible positive integer. Its convergents are ratios of consecutive Fibonacci numbers:

$$1,\ \frac{2}{1},\ \frac{3}{2},\ \frac{5}{3},\ \frac{8}{5},\ \frac{13}{8}, \ldots, \qquad \frac{F_{n+1}}{F_n} \to \varphi.$$

In Diophantine approximation, the golden ratio is a canonical maximally badly approximable number: small-denominator rational approximations improve as slowly as possible, and the golden ratio is extremal in Hurwitz's theorem. The phrase "most irrational" refers to this resistance to rational approximation, not to a different logical status among irrational numbers.

The golden angle therefore offers:

(1) irrationality, which prevents exact periodicity;

(2) poor small-denominator rational approximations, which suppress short near-periods.

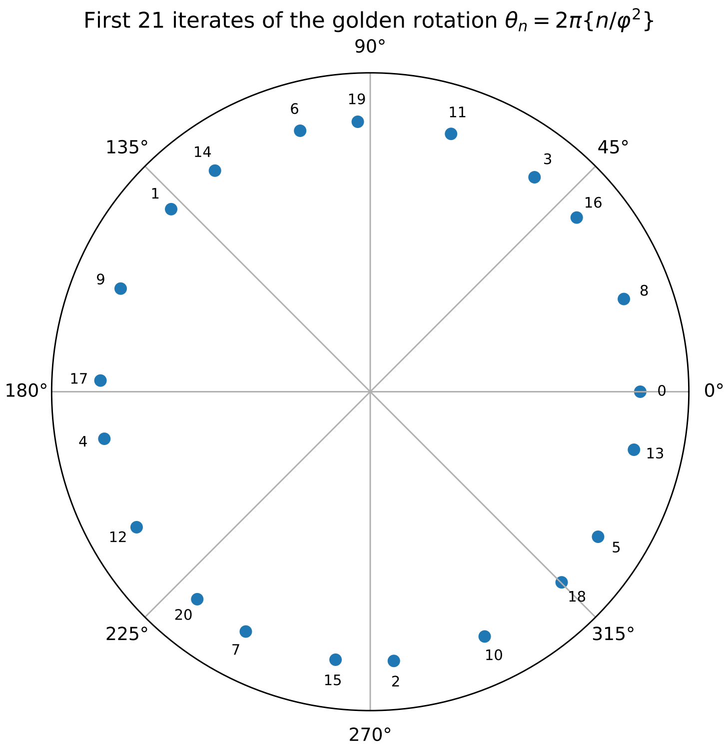

Figure 3: The first 21 iterates of the golden rotation. Labels show generation order. The golden angle does not place every new node exactly at the centre of the current largest gap, but it avoids low-order resonance and gives stable finite prefixes.

8.4 The three-gap theorem as supplementary intuition

For any irrational rotation, the first N iterates divide the circle into gaps of at most three distinct lengths. This is the three-gap theorem. The theorem is not specific to the golden ratio, but the Fibonacci continued-fraction structure of the golden rotation gives especially balanced gap hierarchies at many Fibonacci-sized prefixes.

One should not overstate the conclusion. The golden angle is not guaranteed to be the exact optimal packing for every finite N, nor does the standard spherical Fibonacci formula necessarily minimise the mesh ratio for every N. It is attractive because it is generated in $O(N)$ time, is nearly equal-area, avoids periodic anisotropy, and performs very well in separation, covering, discrepancy, and practical integration tests. Modified Fibonacci variants can improve polar separation constants[4].

9 Relation to the 0.618 method in operations research

The same golden ratio occurs, but for a different optimisation reason.

9.1 Why golden-section search uses 0.618

Normalise the current interval to $[0, 1]$ and place two interior points at

$$c = 1 - r, \qquad d = r, \qquad r > \frac{1}{2}.$$

After comparing function values, suppose the left subinterval $[0, r]$ is retained. To make the old point $c = 1 - r$ occupy the same relative golden position in the new interval–so that one function value can be reused–we require

$$1 - r = r^2.$$

The positive solution is

$$r = \frac{\sqrt{5} - 1}{2} = \frac{1}{\varphi} \approx 0.618034.$$

Thus each iteration preserves a self-similar interval geometry and needs only one new function evaluation. Golden-section search is the limiting form of Fibonacci search as the allowed number of evaluations grows; the minimax construction goes back to Kiefer[11].

9.2 Same algebraic constant, different mechanisms

The golden angle uses $1/\varphi^2 \approx 0.381966$ of a turn, while golden-section search uses $1/\varphi \approx 0.618034$. They are complementary:

$$\frac{1}{\varphi} + \frac{1}{\varphi^2} = 1.$$

Modulo one turn, a forward rotation by $1/\varphi^2$ is equivalent to a backward rotation by $1/\varphi$; the same circle positions are generated in the opposite direction.

Connection and distinction

- Golden angle: suppress small-denominator near-periods in a rotation sequence.
- Golden-section search: preserve self-similarity of a shrinking interval and reuse one evaluation.
- Both arise from $\varphi^2 = \varphi + 1$, but the algorithmic roles are different.

10 Finite geometric comparison: random versus Fibonacci points

10.1 Nearest-neighbour distances

Figure 4 compares nearest-neighbour geodesic distances for 3000 i.i.d. random points and 3000 Fibonacci points. The random distribution has a long left tail, representing unusually close pairs. Fibonacci nearest-neighbour distances are concentrated in a narrow range, reflecting stable local spacing.

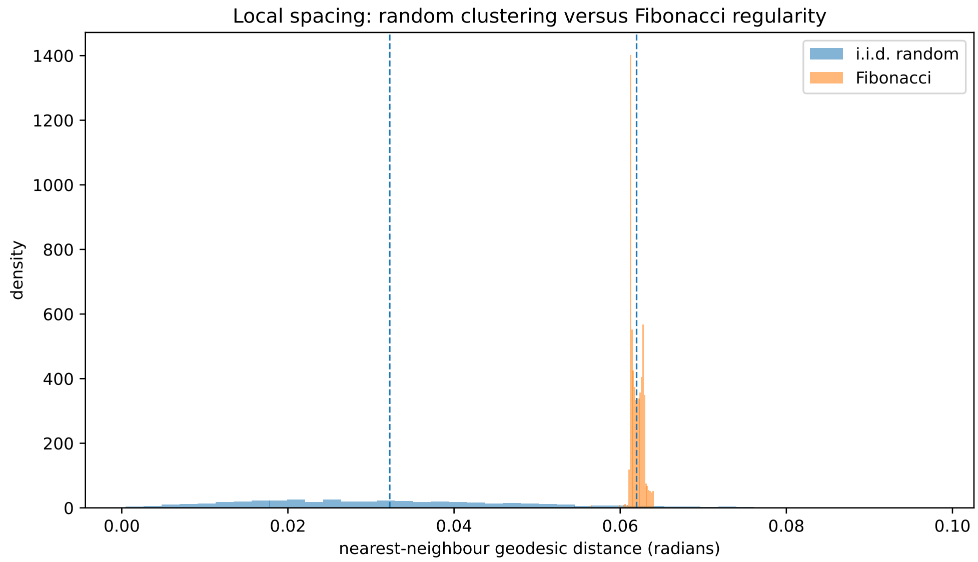

Figure 4: Nearest-neighbour distances. Random points permit clustering; Fibonacci points behave like a deterministic repulsive layout. The plot is an illustration, not a substitute for asymptotic theorems.

10.2 Separation, covering, and mesh ratio

Figure 5 estimates the covering radius with a dense test grid. Fibonacci separation and covering both follow an $N^{-1/2}$ trend. For random points, the minimum separation is controlled by extreme close pairs and is nearer an N^{-1} scale, while covering is worsened by extreme holes. Figure 6 consequently shows a roughly stable Fibonacci mesh ratio and a growing random mesh ratio. Since the covering radius is numerically approximated, these figures should be interpreted as diagnostics rather than proofs.

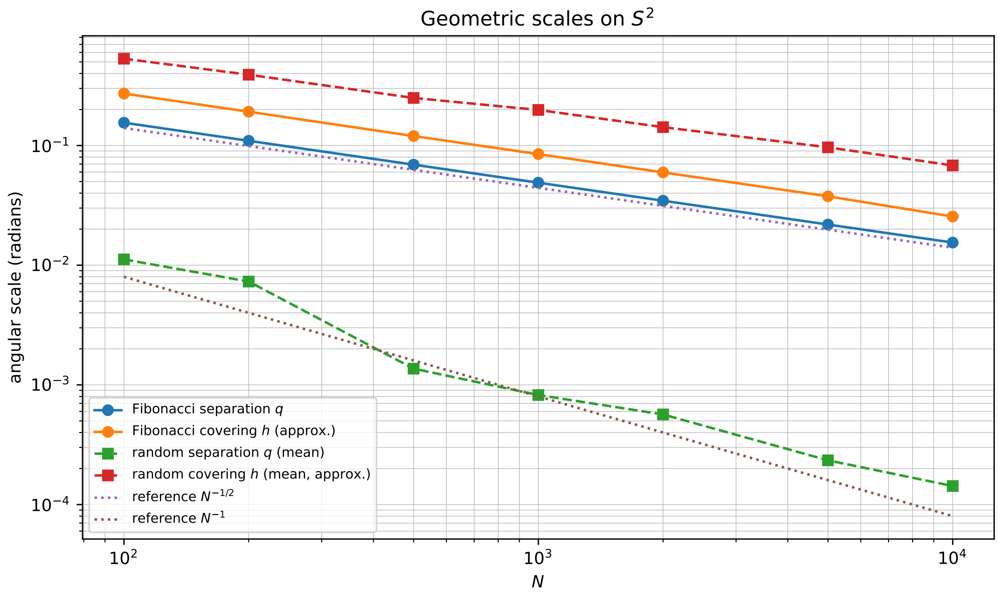

Figure 5: Computed geometric scales on $\mathbb{S}^2$. Fibonacci separation and coverage both track $N^{-1/2}$; random minimum separation is dominated by extreme close pairs.

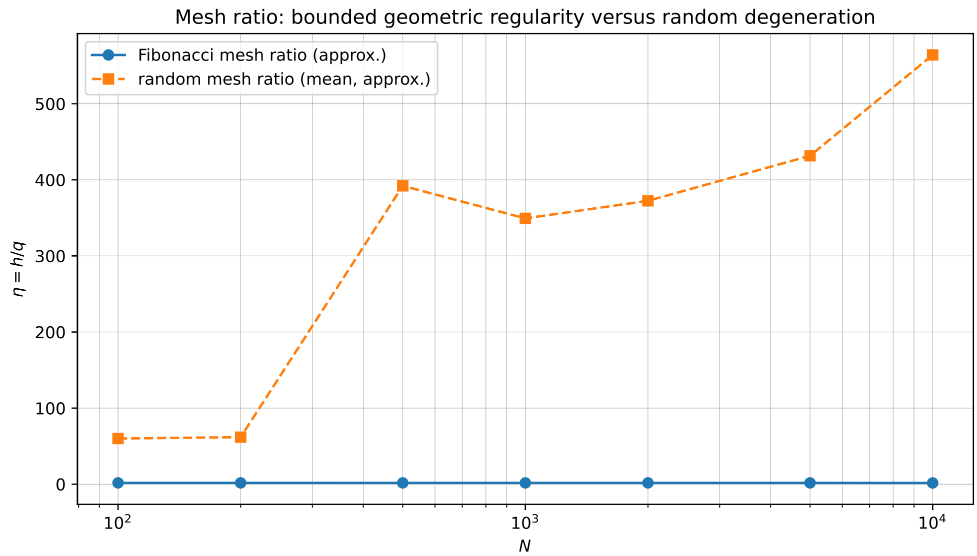

Figure 6: Approximate mesh ratios. Fibonacci configurations retain geometric shape regularity, whereas finite i.i.d. configurations degenerate in mesh ratio despite empirical equidistribution.

10.3 Discrepancy is a third notion of uniformity

For a spherical cap C, define the spherical-cap discrepancy

$$D_N = \sup_C \left| \frac{1}{N} \#(X_N \cap C) - \mu(C) \right|.$$

This measures the worst difference between empirical point proportion and true area proportion. Discrepancy and mesh ratio address different features:

- mesh ratio is local-extreme geometry: closest pairs and largest holes;
- discrepancy is counting error over a family of test sets;
- integration error also depends on function smoothness and the spectral or design properties of the nodes.

Spherical Fibonacci lattices have been studied for spherical-cap discrepancy and area measurement[8, 7]. Thus the statement "Fibonacci points are more uniform" should be decomposed into precise claims: near equal-area allocation, stable local spacing, good coverage, and small error for many spherical counting and integration tasks.

11 Interface with convex geometry and the current code

Convex geometry is the application context. Support functions, radial functions, and spherical integration are basic tools in constant-width geometry; see Section 2.4 and Chapter 13 of Martini–Montejano–Oliveros and Schneider's Brunn–Minkowski monograph[1, 2].

11.1 Why quasi-uniform support directions matter

For a convex body $K \subset \mathbb{R}^3$,

$$K = \bigcap_{\theta \in \mathbb{S}^2} \{x : \langle x, \theta \rangle \le h_K(\theta)\}.$$

Keeping only directions $\theta_1, \ldots, \theta_m$ gives the outer approximation

$$K_m = \bigcap_{j=1}^{m} \{x : \langle x, \theta_j \rangle \le h_K(\theta_j)\}, \qquad K \subseteq K_m.$$

If the direction set has a large hole, the genuinely active supporting direction may be missed and the approximation can bulge outward in that region. The error is driven by the worst omitted direction, so covering radius is more relevant than correctness of an average distribution.

For the current U_3 code,

$$\rho_{U_3}(u) = \inf_{\langle u, \theta \rangle > 0} \frac{h_{U_3}(\theta)}{\langle u, \theta \rangle}.$$

After discretisation, the infimum is replaced by a finite minimum. Missing the true minimising direction overestimates the radial value. Because the program repeatedly computes inf, min, sup, and max, worst-case holes are more important than average sample balance.

11.2 What exactly is discretised?

Configuration or parameter	Space	Typical size	Role

Meissner arc nodes	three one-dimensional arcs	3×4096	Replace a continuous family of generating ball centres
Meissner base directions	$\mathbb{S}^2$	200000	Shoot radial rays and sample the three-dimensional boundary
U_3 support directions	$\mathbb{S}^2$	100000	Discretise the continuum of supporting halfspaces
U_3 base directions	$\mathbb{S}^2$	100000	Sample the radial boundary
Radial layers	$K_0 \subset \mathbb{R}^3$	9 per direction	Sample the entire base body, not only its boundary
Upper-graph centre cloud	$\mathbb{R}^4$	about 0.9–1.8 million	Centres of unit four-balls defining the lower raised half
Final integration directions	$\mathbb{S}^3$	20000 antipodal pairs	i.i.d. normalised Gaussian directions for radial Monte Carlo

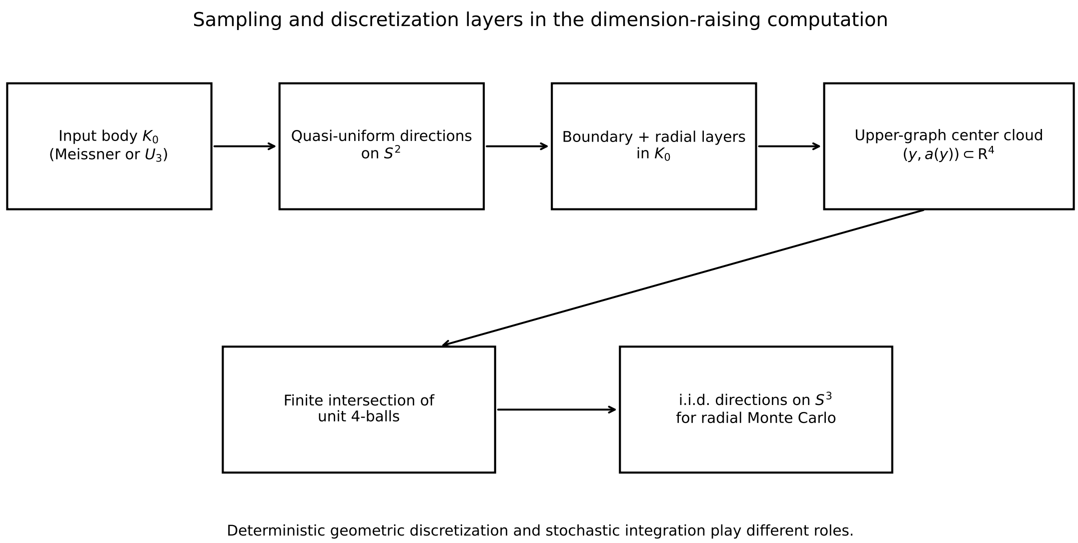

Figure 7: Sampling layers in the dimension-raising computation. Fibonacci points on $\mathbb{S}^2$ describe geometry; i.i.d. points on $\mathbb{S}^3$ perform the statistical integral.

11.3 What is an upper-graph centre cloud?

For a three-dimensional input body K_0, define the upper height

$$a(y) = \sqrt{1 - d_0(y)^2}, \qquad d_0(y) = \max_{q \in K_0} \|y - q\|.$$

The continuous upper graph is

$$G_{K_0} = \{(y, a(y)) : y \in K_0\} \subset \mathbb{R}^4.$$

The lower half of the raised body can be represented as an intersection of unit four-balls centred on this graph. The code first samples boundary directions quasi-uniformly, then takes several radial layers to obtain points $y_{i,\ell}$ throughout K_0, and lifts them to

$$c_{i,\ell} = (y_{i,\ell}, a(y_{i,\ell})) \in \mathbb{R}^4.$$

There is no explicit triangulation; the data are simply a large collection of coordinates, hence a point cloud. Since each point is used as the centre of a unit four-ball, it is a centre cloud.

11.4 Why are the final integration directions still random?

The final estimator samples i.i.d. normalised Gaussian directions on $\mathbb{S}^3$ and computes

$$\widehat{V}_N = \frac{\pi^2}{2N} \sum_{j=1}^{N} \rho_K(U_j)^4.$$

At this stage standard errors, confidence intervals, antithetic pairs, and paired body differences are required. Independence is therefore valuable. A deterministic quasi-Monte Carlo rule might reduce error, but ordinary i.i.d. standard-error formulas would no longer apply; one would need randomised QMC, repeated random rotations, or worst-case error analysis.

The present design is therefore intentional:

geometric discretisation:	Fibonacci / quasi-uniform nodes,
statistical integration:	i.i.d. uniform random directions.

12 How should spherical point sets be assessed and selected?

12.1 Recommended diagnostics

For direction sets used in convex-body approximation, one should report or monitor:

(1) separation radius q;

(2) covering radius h;

(3) mesh ratio h/q across resolutions;

(4) spherical-cap discrepancy or another equal-area diagnostic;

(5) a problem-specific resolution study: how much does the volume or extremum change when the number of directions is doubled?

12.2 Different configurations serve different purposes

- **i.i.d. random points**: simplest and ideal for classical Monte Carlo error analysis, but inefficient as a geometric mesh.
- **Fibonacci points**: $O(N)$ generation, near equal-area allocation, low anisotropy, and strong practical separation/coverage; suitable for very large support and boundary direction sets.

- **spherical designs**: exact for low-degree spherical polynomials and highly effective for smooth quadrature, but often more expensive to construct.
- **energy-minimising or maximin points**: excellent separation and coverage, but optimisation costs can be much larger.
- **latitude–longitude grids**: convenient indexing, but polar clustering and anisotropy make them poor equal-area meshes.

12.3 A warning about $\mathbb{S}^3$

The formula in `fibonacci_sphere` exploits the special (z, ϕ) area coordinates of $\mathbb{S}^2$ and does not transfer unchanged to $\mathbb{S}^3$. On $\mathbb{S}^3$, the natural geometric spacing is $N^{-1/3}$. Possible alternatives include normalised low-discrepancy sequences, Hopf-coordinate constructions, spherical designs, energy points, or randomised QMC. The current i.i.d. choice on $\mathbb{S}^3$ is statistically robust; randomised QMC becomes relevant if integration variance, rather than geometric discretisation, becomes the dominant bottleneck.

13 Conclusion and memory framework

One-page summary

1. Probability uniformity means that every point has the correct surface-area distribution; geometric uniformity means that the finite configuration has neither severe clusters nor large holes.

2. Separation radius
$$q = \frac{1}{2}\min_{i\neq j} d(x_i, x_j)$$
controls clustering, while covering radius
$$h = \sup_x \min_i d(x, x_i)$$
controls holes.

3. Mesh ratio
$$\eta = h/q$$
is a dimensionless shape-regularity measure. Quasi-uniformity requires $\eta \leq C$ with C independent of N.

4. On a d-dimensional manifold, the natural geometric spacing is $N^{-1/d}$. On $\mathbb{S}^2$ it is $N^{-1/2}$. This is unrelated to the Monte Carlo $N^{-1/2}$ rate except for the coincident exponent.

5. In the Fibonacci construction,
$$z_i = 1 - \frac{2(i+1/2)}{N}$$
selects midpoints of equal-area horizontal zones, while
$$\phi_i = i\pi(3-\sqrt{5})$$
uses the golden angle to suppress longitudinal periodicity.

6. The golden angle and the 0.618 optimisation method both derive from $\varphi^2 = \varphi + 1$, but one addresses irrational-rotation resonance and the other interval self-similarity and function-value reuse.

7. In the present code, Fibonacci nodes discretise support and boundary directions on $\mathbb{S}^2$; final integration on $\mathbb{S}^3$ uses i.i.d. random directions.

A Terminology

Term	Meaning
uniform distribution	Probability law equal to normalised surface-area measure

equidistribution	Empirical proportions converge to area proportions
separation radius	Half the minimum pairwise distance
covering radius / fill distance	Radius of the largest hole
mesh ratio	Covering radius divided by separation radius
quasi-uniform	Mesh ratio bounded independently of N
spherical-cap discrepancy	Worst counting error over spherical caps
golden angle	$2\pi/\varphi^2 \approx 137.5°$
irrational rotation	Circle rotation by an irrational fraction of a turn
centre cloud	Finite high-dimensional set used as ball centres
radial direction	Direction of a ray from a fixed interior point
support direction	Unit normal to a supporting hyperplane

B Line-by-line mathematical interpretation of the code

The five essential lines mean

$$\begin{aligned}
i &= 0,\dots,N-1 && \text{index equal-area zones,}\\
z_i &= 1-2(i+1/2)/N && \text{choose the midpoint height of zone } i,\\
\phi_i &= i\pi(3-\sqrt{5}) && \text{advance longitude by the golden angle,}\\
r_i &= \sqrt{1-z_i^2} && \text{compute the latitude-circle radius,}\\
x_i &= (r_i\cos\phi_i, r_i\sin\phi_i, z_i) && \text{obtain a point on the sphere.}
\end{aligned}$$

The shortest useful mnemonic is

$$\boxed{\text{equal area from } z_i, \qquad \text{stripe avoidance from the golden angle.}}$$

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